

CHEST PATCH AND UPPER-BODY CONTEXT EMBODIMENT — SYSTEMIC PHYSIOLOGY IMPLEMENTATION

CHEST-MOUNTED IMPLEMENTATION

In one embodiment, Node 5 may be implemented as a chest-mounted physiological monitoring device configured to acquire systemic biometric signals associated with full-body contextual interpretation of intimate physiology. Placement over the thoracic region enables acquisition of cardiac, respiratory, autonomic, and motion-related signals representing central physiological state information.

DEVICE STRUCTURE

The chest embodiment may comprise an adhesive patch, conformal wearable module, textile-integrated sensor platform, strap-mounted device, magnetic attachment system, or hybrid mounting architecture. The device housing may include flexible polymeric substrates, silicone encapsulation layers, fabric integration structures, or semi-rigid enclosures configured for long-duration comfortable wear.

PHYSIOLOGICAL SIGNAL ACQUISITION

Positioning on or near the sternum allows collection of high-quality cardiovascular and respiratory signals. The device may incorporate electrocardiography electrodes, photoplethysmography sensors, bioimpedance measurement circuits, respiration expansion sensors, skin temperature sensors, electrodermal activity sensors, accelerometers, gyroscopes, and environmental sensing components to characterize systemic physiological state.

CONTEXTUAL REFERENCE FUNCTION

The chest-mounted embodiment functions as a central reference node capturing autonomic nervous system activity, emotional engagement indicators, physical exertion levels, stress responses, and recovery dynamics. These signals provide interpretive context for localized intimate physiology measurements obtained from additional nodes within the distributed system.

POWER AND COMMUNICATION ARCHITECTURE

The device may utilize rechargeable batteries, thin-film power cells, inductive charging systems, energy harvesting mechanisms, or external power modules. Communication with companion nodes may occur via wireless protocols including Bluetooth Low Energy, ultra-wideband communication, Wi-Fi, near-field communication, or future wireless standards supporting synchronized physiological data exchange.

ALTERNATIVE UPPER-BODY FORM FACTORS

Alternative embodiments may include necklace-mounted sensors, clavicle-mounted modules, shoulder wearables, garment-integrated biosensing systems, or distributed multi-patch configurations operating cooperatively. Any body-positioned system capable of capturing systemic physiological context shall be considered within the scope of the disclosed Node 5 architecture.